Technical Report

Optimisation Model for Investigating Fraudulent Transactions in the Banking Sector

Daniel Kopisitskiy, Nicolas Talia, Teng Wei Yeo, Sara Brdnik



THE UNIVERSITY *of* EDINBURGH

Methodology, Modelling and Consulting Skills
University of Edinburgh
School of Mathematics
November 2023

Executive Summary

Frauds are a significant cost to the banking sector, given new regulations which mandates banks to compensate scammed customers. This report presents **two** optimisation models which learn from past data to advise banks how to choose which transactions to investigate in future days.

We have two optimisation models that satisfy different needs. Our **first model** is **better at protecting customers** and ensuring their trust, because a **higher proportion of scams are caught**. However, the first model sacrifices on costs, as it hires many external investigators. Hence, we constructed a **second model** which prioritises **maximising the transaction value of scams caught**, while **minimising hiring costs**. Hence, if the regulator prioritises catching frauds, it should use the first model. If the regulator prioritises costs, it should use the second model.

The similarities of both models are as follows: both models feature **two stages**. In the **first stage** both models learn from **past data** by identifying features of transactions (category, transaction probability, day of the month, etc.) which are most relevant to predicting whether a transaction is a fraud or not. More relevant features are given bigger weights. In the **second stage**, both models use the weights found from the first stage to compute **probabilities** a transaction should be investigated for each new transaction on the next day. This probability is then used to generate a set of **candidate cases** which the bank should investigate, then use **linear programming** to confirm whether these candidate cases are **worth investigating** when considering operational costs and investigation time. If a candidate case requires an external investigator, and it costs more to investigate than its transaction amount, then the models will inform you that it is not worth investigating.

The differences between the models are as follows: the computational time of the first model is better, which makes it more scaleable to larger datasets. The first model also analyses more features of each transaction to detect fraud.

Another subtle difference is that the first model transforms the probabilities found in the second stage into candidate cases using a **threshold**; cases with a computed probability of being a scam higher than 0.03 will be a candidate case. However, the second model does not use a threshold. Instead, it **uses these probabilities** as a measure of our **uncertainty**. Our second model samples candidate cases **using these probabilities**. A case which has a low computed probability of being fraud will be unlikely to be suggested as a candidate case, while a case which has a high computed probability of being fraudulent will be more likely to be suggested as a candidate case.

A crucial strength of the second model is that it can be **applied to real world data where banks can only see the results of previously-investigated cases**; our second model works even with **no information about uninvestigated cases**. This is due to its probabilistic approach; the probability a case is a fraud represents the uncertainty that it is a fraud. Even if the model thinks a case has low probability of being a fraud, our model *might* still suggest it to be a candidate case. This allows the model to be flexible to changes in the dataset, as it continues to investigate a wide range of cases, and learn from these investigated cases. Model results are in Table 1.

	Model 1	**Model 2**	**Model 2 without knowing false negatives**	
			Average	**Std.Dev.**
Theoretical max loss £	− 350 328.25	− 350 328.25	− 350 328.25	
Our detected scams £	+ 314 369.50	+ 298 387.10	+ 263 118.02	42 172.92
Hiring cost £	− 50 950.00	− 1 900.00	− 75 440.00	12 963.59
Our actual loss £	− 86 908.75	− 53 841.15	− 162 580.23	54 565.05
Proportion of losses **cut**	**75.19%**	**84.63%**	**53.59%**	15.58%

Table 1: Balance sheet for both models, and for model 2 without seeing false negatives. The average and standard deviation for Model 2 (without knowing false negatives) is computed based on 5 trials. Theoretical max loss is the maximum loss from fraud if no investigations were done.

For elaboration on model comparisons, the application of the fully-trained models on the full data set, the model implementation in the case that false negatives are not known, and future model improvements, see the technical report. A link to our code can be found at https://github.com/nicolastalia/mmcs_final.git. The repository is private; email for access.

Contents

1 Introduction **1**

2 Problem **1**
- 2.1 Problem Outline . . . 1
- 2.2 Programming and Data Cleaning . . . 2
- 2.3 Modelling Motivations . . . 2

3 Model 1 **2**
- 3.1 Assumptions . . . 2
- 3.2 Model 1, Stage 1 . . . 2
- 3.3 Model 1, Stage 2 . . . 3
- 3.4 Results . . . 4

4 Model 2 **4**
- 4.1 Assumptions and Outline . . . 4
- 4.2 Data Insights . . . 4
- 4.3 Model 2, Stage 1 - Determining Weights . . . 5
- 4.4 Model 2, Stage 2 - Finding Candidate Cases and Allocating Investigators . . . 6
- 4.5 Results . . . 7

5 Discussion and Extensions **8**
- 5.1 Comparing Models 1 and 2 . . . 8
- 5.2 Applying Trained Model 2 to Full Dataset . . . 8
- 5.3 No False Negatives - Model 2 . . . 9

6 Future Model Improvements **10**

7 Conclusion **10**

References **11**

A Appendix **12**
- A.1 Model 2 Stage 1 . . . 12
- A.2 Model 2 - Example Results . . . 12
- A.3 Data Analysis: Scams by Category . . . 13
- A.4 Nomenclature Model 1 . . . 14
- A.5 Nomenclature Model 2 . . . 15

1 Introduction

Money lost from financial scams reached £1.2 billion in 2022 [1]. There is hence an urgent need for fraud detection strategies. Furthermore, new legislation in 2024 will require both receiving and sending banks to refund fraudulent transactions, incentivising inter-bank cooperation [2]. Such costs are to be spit 50:50 between the two banks [3]. While bearing in mind the importance of minimising the loss from frauds, the Payment Systems Regulator further stresses the significance of increasing protection, trust and security of customers [3]. As the consumer and fraudster behaviour changes fast, there is exigency in finding the key identifiers of fraud that can be detected by a mathematical model [4].

This project pertains the loss of money from financial scams and aims to assist the banks' decision to investigate transactions to minimise loss. As a small-scale simulation of the UK banking system, the report includes two models that aim to identify fraudulent cases, and gives advice to the decision maker on which cases to investigate, given the banks' goals.

The first model uses multivariate linear regression (MLR) to create classification of a transaction. The second model combines the key identifiers into a value that presents the probability that the transaction should be investigated, and it then uses the Bernoulli distribution to sample the transactions that are to be investigated (a Bernoulli distribution is a Binomial with one trial). Both models have two stages and the main focus was on the developing the first stage as the second stages are similar for both models and mostly just operational.

Section 2 outlines the problem and some information about the datasets. Section 3 explains the first model, its assumptions, and results. The second model and its results are explained in Section 4. Section 5 discusses extensions to the second model, particularly the case of no-false-negatives. The next section presents the possible improvements to the models and the last chapter concludes.

2 Problem

2.1 Problem Outline

This section presents the specific problem set-up of the project [5]. The transactions in this model originate from one of 5 banks (A, B, C, D and E). The payments could be made to one of the 5 banks, including within a bank; to an unknown international bank; or the recipient could be unknown (which is the case for ATM withdrawals). Table 2 outlines the number of internal investigators available at each bank.

Each case is associated with customer, description and transactional probabilities. Customer probability is constant for each user and depends on their past behaviour and likelihood of being scammed. Transaction and description probabilities are unique to each transaction and are related to the financial component of the transaction, determined internally by each bank or to their descriptions, respectively. Furthermore, there is information about the category of a given transaction (eg. shopping, investment), its amount, and the time of the month during which the funds were sent.

Table 2: Bank Investigation Team Size

Bank	**A**	**B**	**C**	**D**	**E**
Team Size	8	12	10	10	10

Table 3: Resources for Investigation by Priority

Priority	1	2	3	4
Time Spent (days)	0.25	0.5	1	2
External Cost (£)	40	60	100	150

Every case has one of 4 levels of priorities. If the analysis is conducted by an internal team member, its investigation time depends on the priority. A bank can additionally employ external investigators whose service charges are based on the priority of the case (Table 3). As per the new legislation [3], such costs are split equally between the two banks included in the transaction, unless the transaction is internal or international. Then, only the domestic sending bank pays for the costs.

2.2 Programming and Data Cleaning

Python Xpress is used throughout. Cash withdrawal entries and money paid in (income, interests from savings accounts) were removed during preprocessing as they do not tend to be of interest to the banks. It is assumed that all withdrawals are genuine.

2.3 Modelling Motivations

There are two models in this report, each consisting of two stages: (1) minimise expected loss, and (2) use the knowledge gained from stage 1 to find candidate cases. Then an optimisation model is used to decide whether these cases are worth investigating and how to distribute them [6].

Let us call $p_i = \mathbf{w} \cdot \mathbf{x}$ the probability of investigating transaction i with features $\mathbf{x}$. In the first stage the decision variables was obtained $\mathbf{w}$ via

$$\min \sum_i f(p_i, \text{ features of transaction i}) \tag{A}$$

Then the weights $\mathbf{w}$ were applied on the next day to find that day's candidate transactions to investigate, and distribute investigation capacities such that the expected loss is minimal.

The distinction between models (1) and (2) is in the definition of f. Model (1) is a classification model which adjusts probabilities that *a transaction is a fraud*. Model (2) is a model which aggregates criteria to find probabilities *that a case should be investigated*. The distinction is important because model 2 models for time constraints and hiring costs in the first optimisation stage, but model 1 does not.

Another distinction is that model (1) implements a threshold for deciding whether an investigation is worth investigating, whereas model (2) uses a probabilistic approach. It gives a *probability* that the bank should investigate a particular case.

We have two models because they serve different purposes. Model 1 is better at detecting frauds (true positives) and is computationally more efficient. But model 2 saves more money, and it can provide decisions even when false negatives are not known to the bank.

3 Model 1

3.1 Assumptions

For model 1, we assume the following: only banks involved in a transaction can investigate it - either by a person from their team or by hiring an external investigator. Priority 4 transactions take 2 days, which was simplified by allocating them 2 investigators to complete the analysis within a day.

As the costs of refunds will be split between the two banks equally starting in 2024 [2], this model assumes a 50:50 split of the costs for investigating transactions if both banks agree to investigate. There are three situations where only the bank where the transaction originates from pays for the costs: the other bank is not interested in investigating, the receiving bank is international or it is the same as the sending bank.

We assume that the weights found based on the perfect information from the previous days fit the cases on the next day. Refer to appendix A.4 for definitions of variables.

3.2 Model 1, Stage 1

This model uses multivariate linear regression to classify the transactions into fraud and non-fraud.

s_i is a binary value that equals 1 if it is known that the transaction i was a fraud and 0 otherwise. $\hat{s}_i$ is its predicted value, the probability that case i is a fraud. Then, the best-fit line with intercept m is found by minimizing the residual sum of squares.

$\mathbf{x_i^b}$ is the vector associated with the banks $\{A, B, C, D, E\}$ and it depends on which bank the funds are sent from. Since this is a categorical attribute, the vector assigns a value of 1 if the transaction was done from bank b, and 0 otherwise.

$\mathbf{x_i^p}$, $\mathbf{x_i^c}$, $\mathbf{x_i^d}$ are vectors that assign 1 if the priority, category, and day equal to the one in the transaction i, and zero for the rest of the elements of the vector. This allows the model to have a weight based on each priority, category and day of the month. tp_i, dp_i and cp_i represent transactional, description and customer probability for each case i, respectively.

Each indicator that were just explained gets its associated weight so as to find the best fitted line. Their respective weights are denoted as $\mathbf{w^b}$, $\mathbf{w^p}$, $\mathbf{w^c}$, $\mathbf{w^d}$, w^{tp}, w^{dp}, w^{cp}. The predicted value of whether the transaction is a scam or not is defined as:

$$\hat{s} = m + \mathbf{w^b} \cdot \mathbf{x_i^b} + \mathbf{w^p} \cdot \mathbf{x_i^p} + \mathbf{w^c} \cdot \mathbf{x_i^c} + \mathbf{w^d} \cdot \mathbf{x_i^d} + w^{tp} \cdot tp_i + w^{dp} \cdot dp_i + w^{cp} \cdot cp_i \tag{1}$$

Then, to find the best fit line, the sum of the squared differences is minimized:

$$min \quad \sum_i (s_i - \hat{s})^2 \tag{2}$$

3.3 Model 1, Stage 2

In the second stage, the goal is to maximise the amount of money saved by the banks. That equals the amount saved in the transaction a_i, multiplied by the adjusted probability of being a scam q_i, found in Stage 1, minus the cost of hiring an external investigator for this transaction $E_p * y_{i,bf}$. The value $x_{bf,bt}$ equals 1 if the case should be investigated and 0 otherwise. The value $y_{bf,bt}$ equals 1 if the case should hire an external investigator, and 0 otherwise.

By taking into account the costs of the external investigators, it guarantees that the low-value transactions are not prioritised, even if they are very likely to be fraudulent.

$$max \quad \sum_i \left[a_i \cdot q_i \cdot x_{i,bf} - y_{i,bf} \cdot E_p \right]$$

$$q_i = \begin{cases} 1 & \text{if } \hat{s} \geq 0.8 \\ tp_i & \text{if } 0.03 \leq \hat{s} \leq 0.8 \\ 0 & \text{if } \hat{s} < 0.3 \end{cases} \qquad \forall i \in I \tag{3}$$

To ensure the model is not too deterministic, the probabilities q_i are adjusted so value 1 is given to the cases that are most likely to be fraud, and 0 are given to the cases that are very unlikely to be frauds (value ≤ 0.03). The rest of the values are assigned the value of their transaction probability, tp_i as given in the data. Values 0.8 and 0.03 were determined by analyzing the training dataset after running stage 1.

$$\sum_i (u_i \cdot T_i \cdot (x_{i,bf} - y_{i,bf})) + \frac{1}{2} \sum_i \left[(1 - u_i) \cdot T_i \cdot \left[(x_{i,bf} - y_{i,bf}) + (x_{i,bt} - y_{i,bt}) \right] \right] \leq Z_b \tag{C1}$$

$$x_{bf,t} + x_{bt,t} = 2 \cdot z \tag{C2}$$

$$y_{bf,t} = y_{bt,t} \tag{C3}$$

$$y_{bf,t} \leq x_{bf,t} \tag{C4}$$

$$y_{bt,t} \leq x_{bt,t} \tag{C5}$$

$$x_{bf,t}, x_{bt,t}, y_{bf,t}, y_{bt,t}, z_t \in \{0, 1\} \tag{C6}$$

The first constraint, C1, limits the number of people that are working on a case internally to the size of the team for each bank. If the investigation will be done by one bank (because the recipient is an international bank or the bank itself), u_i's value is 1, otherwise 0. If the bank is working on a case themselves, they will spend all the required time on it. Otherwise, the required time is halved for the bank b. The required time T_i depends on the case's priority, which is based on the particular case, hence the subscript was simplified to i. C2 ensures that if there are two banks involved in a transaction, then both or neither must investigate it and C3 ensures that external investigators are shared 50:50 between banks. C4 and C5 are to ensure that an external investigator is not hired for a transaction that is not investigated. C6 sets all decision variables as binary.

3.4 Results

The model runs iteratively for each day from 1st Aug 2024 to 31st Dec 2024. Stage 1 of the model trains on the previous 180 days of data.

	Fraud	Not Fraud	Total
Investigated	557	13 606	14 163
Not investigated	449	108 911	109 360
Total	1 006	122 517	123 523

Table 4: Model 1 - Counts of true positives, false positives, false negatives, and true negatives

Theoretical max loss £ (no investigations)	– 350 328.25
Our detected scams £	+ £314 369.50
Hiring cost £	– £50 950.00
Our actual loss £	– £86 908.75
Proportion of losses cut	75.19%

Table 5: Model 1 - Balance sheet

The first measure of success for the first model is the amount of money that the banks save from investigating and preventing fraudulent transactions. The costs in Table 5 include the cost for the external investigators and the costs of having to refund the users whose scams we did not catch. Overall, a significant proportion of money loss is prevented, approximately 75%. Table 4 gives more insights into the specificity and sensitivity of the first model, and Table 6 presents the success rates by priority.

Transaction Priority	Investigation Success Rate
1	48.92%
2	39.81%
3	75.00%
4	69.12%

Table 6: Model 1 - Success rate by priority

4 Model 2

We now wish to develop a linear model in line with a realistic decision making process (by capturing **uncertainty** as encoded by probabilities) of a bank regarding a transaction. Our main measure of success is now the cost we incur from missing fraudulent transactions and hiring investigators.

4.1 Assumptions and Outline

The second model is similar to the first. The main difference is that in the first stage, a *probability* that a case *should be investigated* is computed taking into account also the *expected time constraints* of the banks' investigators. These probabilities are modelled as a weighted average of the different criteria for investigation (categories, amounts, etc.); these weights are found by minimising expected loss given perfect information in the training dataset (all previous days). In the second stage, these probabilities are then used to identify candidate investigations for the subsequent day (where there is no longer perfect information).

Most of the assumptions are similar to the first model, except that the banks are allowed to investigate a case by themselves in the second stage if the other bank does not want to investigate it.

A major assumption is that we assume that weights in the previous day are valid also for the next day. To save on computation time, the seasonality is simplified by splitting a month into 3 parts rather than having a value for each day. It is assumed that the timing impacts the number of frauds (which is confirmed from analysing the training data, as seen the next section).

Variable definitions are in A.5.

4.2 Data Insights

More transactions and frauds happen on the first 4 days of each month than during the days in the rest of the month, as seen in Figure 1. It makes sense that there are more frauds at the start of the month when scammers know that many workers get their income.

However, there is no observable trend of the probability that a transaction is fraudulent within a month. To account for this observation, the values that split month into three parts are introduced; we add three binary variables where ms_i represents the first 4 days, mm_i the days of the month up to the 26th day and me_i the last part of the month.

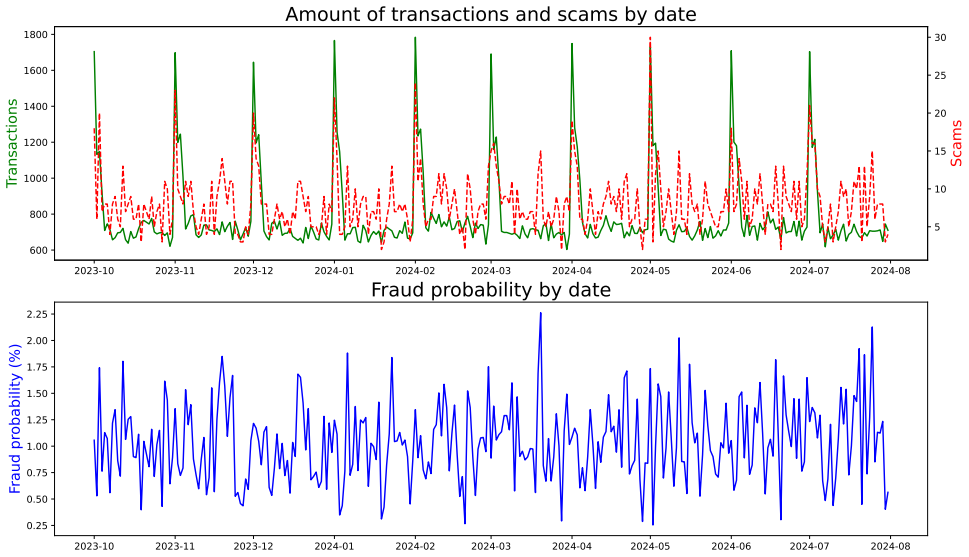


Figure 1: Fraud Data

4.3 Model 2, Stage 1 - Determining Weights

The first stage of the model finds weights which minimise expected loss assuming perfect information from the previous days.

The objective function (4) minimizes the loss from the fraudulent transactions based on the probability that they are not investigated $(1 - combined_i)$ and their amount a_i. That is multiplied by s_i to only get the values from scams since s_i equals 1 if the transaction is a scam and 0 otherwise. The second term takes into account the expected spending for external investigators where $ext_{d,b,i}$ equals the expected number of external investigators and E_i is their cost based on the priority p of the case i.

$$\min \sum_i (1 - \text{combined}_i) \cdot s_i \cdot a_i + \sum_{d,b,i} \text{ext}_{d,b,i} \cdot E_i \tag{4}$$

The choice variables are captured by the value $combined_i$, which tells us the probability that the case i is investigated. The characteristics of a case i that influence the $combined_i$ value are the following:

w^a represents the weight of the amount a_i of a transaction i. This weight was chosen as large transactions should be prioritised so as to not lose lots of funds at once.

w^{tp} and w^{cp} are the weights of transactional probability tp_i and customer probability c_i that are calculated by the banks and given for each transaction. The customer probability is inherent to the sender of the funds. Each transaction has a category and vector $l_{i,c}$ has 1 in the place for category c for a transaction i. Its associated weight is $\boldsymbol{w^c}$, which has weights for each category in their respective index.

To account to variation within a month that was discussed in Section 4.2, a month is separated into three parts and based on the timing of the transaction i, one of the following weights influence the result: w^{start} for the cases in the first 4 days, w^{mid} for the next 22 days and w^{end} for the rest of a month. More weight is given on the weights from the last month than the rest of the data to account for the recent trends. For each transaction i:

$$combined_i = w^a \cdot a_i + w^{tp} \cdot tp_i + w^{cp} \cdot cp_i + \boldsymbol{w^c} \cdot \boldsymbol{l_i^c} + w^{start} \cdot ms_i + w^{mid} \cdot mm_i + w^{end} \cdot me_i \tag{5}$$

Furthermore, $Btime_{i,b}$ is defined as the time it takes for a bank to investigate the transaction i. It is determined by the its priority and the (one or) two bank(s) b included in the transaction. If the

funds were sent between the two out of the 5 banks in the model, $sh_{i,b}$ equals 1 and the time T_i for the sending bank halves. If the recipient is a foreign bank (international) or the bank itself, the time T_i equals to the full time expected for a transaction of the given priority and $so_{i,b}$ equals 1. Otherwise, the case is never investigated and no time is spent on it. $Btime_{i,b}$ is a parameter.

$$Btime_{i,b} = \begin{cases} T_i & \text{if } so_{i,b} = 1 \\ \frac{1}{2}T_i & \text{if } sh_{i,b} = 1 \\ 0 & \text{otherwise} \end{cases} \qquad \forall i \in I \tag{6}$$

The goal is to find how much time each bank is expected to spend for each priority on a given day. The values are separated by priority as the external investigators costs are determined by the priority of the case rather than the time it takes them to investigate it (D1):

$$\exp_{d,b,i} = \sum_{\substack{i \\ \text{s.t. } DD_i=d, \text{PP}_i=p}} \text{combined}_i \cdot \text{Btime}_{i,b} \qquad \forall d \in D, b \in B, p \in P \tag{D1}$$

There is a constraint on the expected time spent on each of the priority of cases, $exp_{d,b,i}$ (D2):

$$\exp_{d,b,i} \leq \text{int}_{d,b,i} + \text{ext}_{d,b,i} \qquad \forall d \in D, b \in B, \ i \text{ s.t. } PP_i = p \tag{D2}$$

The amount of time that internal investigators spend for all cases at a bank equals the sum that they spend for each priority D3. That is true because the time spent for each priority is equal to the proportion of a day an individual investigator would spend for it.

$$\sum_p \text{int}_{d,b,p} = Z_b \qquad \forall b \in B, d \in D \tag{D3}$$

The rest of the constraints ensure that the value of the probability to investigate $combined_i$ and the weights (D5) are non-negative and less than 1. The weight on category $\boldsymbol{w^c}$ is allowed to be negative to increase variety and limited at 0.9 so the model stays linear and does not become too deterministic when certain categories' weights that would equal 1.

$$\text{combined}_i \in [0, 1] \qquad \forall i \in I \tag{D4}$$

$$w^{tp}, w^{cp}, w^{start}, w^{mid}, w^{end}, w^a \in [0, 1] \tag{D5}$$

$$\boldsymbol{w^c} \in (-\infty, 0.9] \qquad \forall c \in C \tag{D6}$$

So, the decision variables of our model are all the weights w, $combined_i$, int, ext, and exp.

4.4 Model 2, Stage 2 - Finding Candidate Cases and Allocating Investigators

In Stage 2, the model uses the weights found in the previous stage and applies them to the new day's dataset. $combined_j$ is calculated for each $j \in J$, where J is the set of transactions for the new day. Then, a list of possible investigation choices X is simulated, where $X_j \sim \text{Bernoulli}(combined_j)$. If $X_j = 0$, then do not investigate case j. Whereas every j for which $X_j = 1$ represents a candidate case for an investigation.

$X_j = 1$ does not imply that case j *will* be investigated; it might not be optimal for the model to investigate case j if it requires an external investigator whose cost exceeds the transaction value of j. Hence, for any set of candidate solutions F such that $X_f = 1$, $f \in J$, a second-phase optimisation model is required, which checks whether these candidate solutions F are indeed worth investigating.

The objective function is minimizing the amount spent for external investigators ($exttwo_f \cdot E_f$) and the amount lost from the fraudulent transactions $(1 - y_f) \cdot a_f$ where $1 - y_f$ represents non-investigated transactions and a_f the amount of such transactions.

The first constraint, E1, ensures that the case y_f is only investigated if there is an external investigator $exttwo_f$ available or if the two banks have available investigators that would work on that case, hence the sum $inttwo_{f,b}$ over all banks b is found. The second constraint E2 ensures that if

$$
\begin{array}{lllll}
\min & \sum_f (exttwo_f \cdot E_f + (1 - y_f) \cdot a_f) & & & \\
\text{s.t.} & exttwo_f + \sum_b inttwo_{f,b} & = & y_f & \forall f \in F \quad \text{(E1)} \\
 & inttwo_{f,b} & \leq & Btime_{f,b} & \forall f \in F, b \in B \quad \text{(E2)} \\
 & \sum_f (inttwo_{f,b} \cdot Btime_{f,b}) & \leq & Z_b & \forall b \in B \quad \text{(E3)}
\end{array}
$$

a bank b is not included in the transaction, they cannot be involved in the investigation of the case. The third constraints ensures that the amount of time that is spent on checking the cases by a bank internally is not greater than the amount of time their team has, Z_b.

The model minimises the total loss summed over all banks and the costs are split for external investigators equally between all banks involved in the relevant transaction. Unlike the first stage of the model, now an unequal split of internal investigation resources between banks involved in a transaction is allowed. To summarise, the model finds weights (on categories, transaction values, customer and transaction probabilities, seasonality) by minimising expected loss in stage 1, then use these weights to construct a probability that an investigator should investigate a case in stage 2.

4.5 Results

The model ran iteratively for each day from August 2024 to December 2024. For the first day of the tests (1st Aug 2024), the model was trained on the cases from October 2023 to July 2024 with perfect information (stage 1); the computed weights are then used to sample for candidate investigation decisions for 1st August 2024 (day one of stage 2). This process is repeated for each day from 1st August 2024 to 31st December 2024; in each iteration, the model trains on the training dataset and all previous days of the testing dataset. The model performance for fraud values is in Table 7 and Figure 2. This model is very successful, as **85% of losses are cut**. The sensitivity and specificity results of the model are in Table 8. Success rates by priority are in Table 9; success rates are not as high as model 1, but the losses cut is greater.

Theoretical max loss £ (no investigations)	– 350 328.25
Our detected scams £	+ 298 387.10
Hiring cost £	– £1 900.00
Our actual loss £	– £53 841.15
Proportion of losses cut	84.63%

Table 7: Model 2 - Balance sheet

	Fraud	Not Fraud	Total
Investigated	440	12 444	12 884
Not investigated	566	110 073	110 639
Total	1 006	122 517	123 523

Table 8: Model 2 - Counts of true positive, false positive, false negative, true negative

Transaction Priority	Investigation Success Rate
1	42.62%
2	12.62%
3	61.21%
4	41.18%

Table 9: Model 2 - Success rate by priority

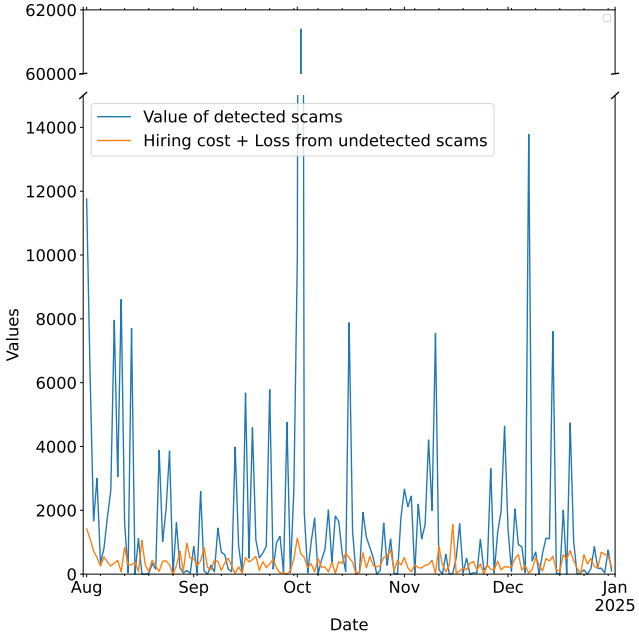


Figure 2: Model 2 - Gains and losses per day

An example of the output from one iteration of training on the training dataset can be found in Appendix A.2. Notably, the model encourages every transaction of certain categories (personal care, shopping, investment, holiday, and entertainment) be investigated; the utilities category adds a 0.05 percentage point to the probability of it being investigated.

The weights for each of the bins for the months converge to the same value by the end of the 5th iteration of testing (i.e. by 5th August 2024), which suggests that the variation in the amount of scam cases by period of month is not significant enough for the model to suggest more cases at the start of the month be investigated.

5 Discussion and Extensions

5.1 Comparing Models 1 and 2

Model 1 is more scalable, because it requires less computation time to train and it catches a higher percentage of true positives (actual fraud). Model 2 achieves better results in the scam value. Though model 2 is linear while model 1 requires minimising a residual sum of squares, they both produce interpretable weights and can be made more flexible by including more features for classification as outlined further below.

5.2 Applying Trained Model 2 to Full Dataset

The success of the fully-trained model is analysed. The weights are firstly optimised by solving Stage 1 with perfect information of the full dataset (training + testing), then the trained model is applied (using the optimised weights) retrospectively to each day of the dataset. This simulates how well the fully trained model were to perform if tested

against each day of the dataset. The results are as follows: Figure 3 and Table 10 shows the value of scams detected. Table 11 sensitivity and specificity. Table 12 shows the success rate by priorities. The target for priority 1 is met, but the model falls short for the rest of the priorities. The losses cut is still high, at 73% of losses cut.

Theoretical max loss £ (no investigations)	– 603 468.65
Our detected scams £	+ 441 592.55
Hiring cost £	– 300.00
Our actual loss £	– 162 176.10
Proportion of losses cut	73.13%

Table 10: Balance sheet from model 1

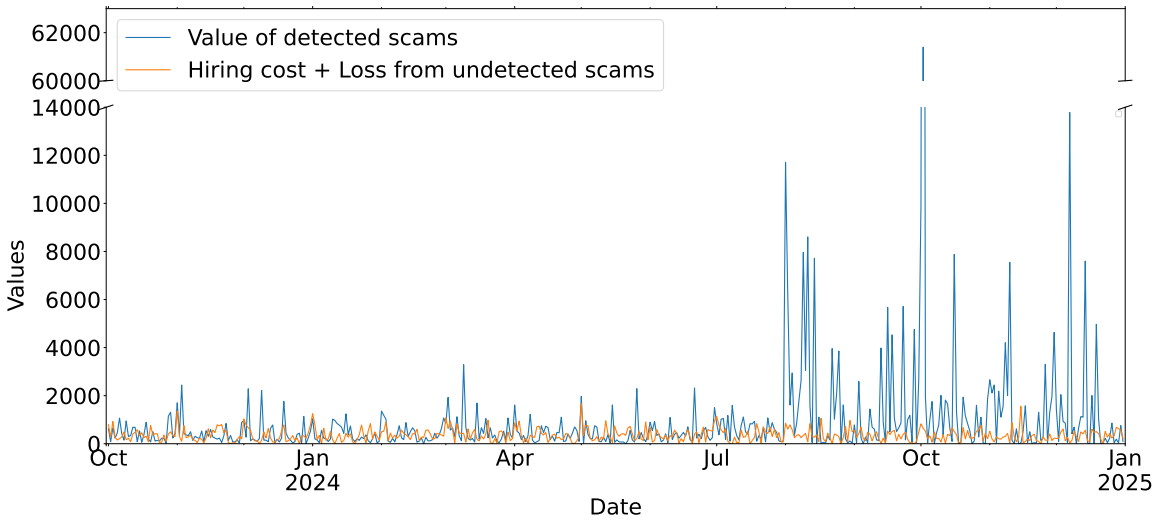


Figure 3: Fully Trained Model (on all data), retroactively applied to all past data

	Fraud	Not Fraud	Total
Investigated	1 706	30 998	32 704
Not investigated	1 781	327 194	328 975
Total	3 487	358 192	361 679

Table 11: Number of true positives, false positives, true negatives, false negatives

Transaction Priority	Investigation Success Rate
1	53.15%
2	27.46%
3	54.22%
4	38.28%

Table 12: Success rate by priority

5.3 No False Negatives - Model 2

The strength of model 2 is that its probabilistic approach lends it well to the real-life problem where banks cannot see information of false negatives. Model 2 is tweaked such that the model can no longer train on the set of all transactions in stage 1; each stage of the model's training now finds the weights which minimises expected loss *only* for the investigations that were *actually* investigated in all previous days. The model can no longer see false (or true) negatives.

Complete ignorance is assumed – the bank starts with having done no prior investigations at all. To model for ignorance, we choose the following uninformative weights for day 0 (1st August 2024) of stage 1: customer weight $w^{cp} = 0.4$, transaction weight $w^{tp} = 0.4$, and 0 for all other weights. Given these initial ignorant weights, the model then proceeds to stage 2; compute $combined_j$ for each $j \in J$, sample $X_j \sim \text{Bernoulli}(combined_j)$. The success of this model is not deterministic, and is dependent on the random choice of which cases it searches on the first day of the testing set to build its weights for future days. As such, 5 trials were calculated to then find the mean and variance of the trials. Table 13 shows the mean and standard deviation of the balance sheet. Figure 4 shows the gain and loss over time. Table 14 shows the specificity and sensitivity. Table 15 shows the investigation success rate by priority.

	Average	Std.Dev.
Theoretical max loss £ (no investigations)	– 350 328.25	
Our detected scams £	+ 263 188.02	42 172.92
Hiring cost £	– 75 440.00	12 963.59
Our actual loss £	– 162 580.23	54 565.05
Proportion of losses cut	53.59%	15.58%

Table 13: Mean and standard deviation of Model 2's balance sheet without seeing false negatives

	Fraud	Not Fraud	Total
Investigated	432.6	14 471	14 903.6
Not Investigated	573.4	108 046	108 619.4
Total	1 006.0	122 517	123 523

Table 14: Specificity and Sensitivity for Model 2 without seeing false negatives (average over 5 runs).

Transaction Priority	Investigation Success Rate
1	38.81%
2	19.81%
3	60.34%
4	56.18%

Table 15: Success rate by priority for Model 2 without seeing false negatives

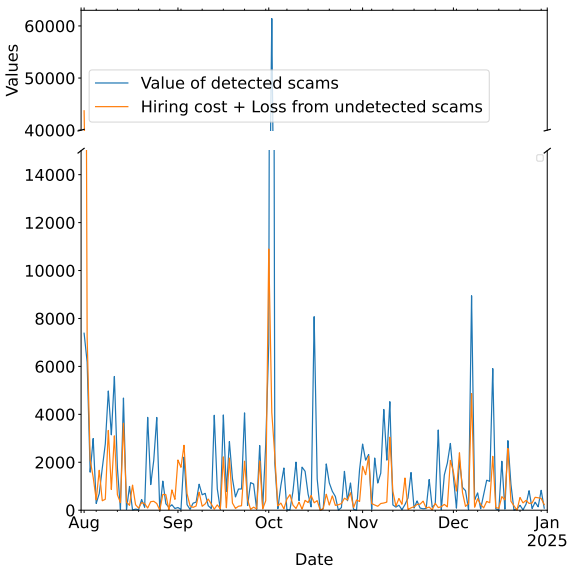


Figure 4: Gain and Loss from Model 2 without seeing false negatives

Even without seeing false negatives, the **model still successfully cuts 54% of losses**. This is the strength of model 2.

6 Future Model Improvements

Our method could be improved of calculating the expected loss in the first stage of the second model by introducing a non-linear constraint on the investigator distribution. Then the expected investigation time could be modelled by taking into account the fact that external investigators are hired for full investigations only and that external investigators do not share investigations with internal ones, but may share investigations with other external investigators. For now, continuous decision variables are used for the external investigation time spent as a shorthand to preserve linearity.

The second model especially is sensitive to parameter tuning. When switching from the test to the train set it quickly converges to the optimal weight distribution. However, this can be further improved by capturing more details of our data set. In particular, the transaction description was not used: machine learning methods for time series analysis could be used, like for example a Long short-term memory model [7], to extract category probabilities instead of the binary values provided in the dataset. This would be in line with past research on detection of financial fraud [8]. Both models' flexibility can be improved by increasing the number of weighted features.

During the second stage of the second model, due to sampling of transactions, the pool of internal investigators is not fully used sometimes. We can remedy that by forcing to investigate the next most likely fraudulent transactions by idle investigators.

Both models 1 and 2 use similar approaches - to derive a series of weights which aggregate evidence over different criteria into a single probability or choice for the decision maker. The models belong to a larger class of methods – aggregation operators – which are used when ranking decisions based on multiple criteria [9]. Complex methods (ordered weighted averages, pairwise comparison matrices) may be explored for further research [10] [11].

Both models also do not account for the fairness across banks; this can be considered in future models.

7 Conclusion

Both models significantly reduce loss compared with doing nothing and catch a substantial portion of fraudulent transactions. The models adapt well to the change in the data from the train to the test set and are particularly good at catching fraudulent transactions that are large in volume.

We make an assumption about the consistency of the data set: in taking past transactions as a predictor of future transactions we base our models on the belief that if there is a correlation between transaction features and fraudulence in the past, there is a high chance of it persisting in the future.

Some of the constraints on the decision boundary in model 1 and on the weights in model 2 are not directly motivated by our model performance and are instead used for reasons of applicability. In particular, the constraint on the category weights in model 2 (to be less than 0.9) reflects the observation that if unconstrained, those weights become 1 for categories particularly prone to fraud, forcing the weights on the transaction volume and probability to be 0. We hence restricted the weights to be less than 0.9.

Another consideration is scalability. Model 2 runs the linear optimisation algorithm on approximately 12,000 decision variables and would benefit from staggering the training window to the most recent 180 days.

One possible suggestion to the banks A to E would be to renegotiate the prices of external investigators, especially priority 2 cases. As it stands right now, a lot of low-value fraud is not investigated because the fraud volume is less than the investigation price.

References

[1] UK Finance Press Release, Feedzai, (2023), "Over £1.2 Billion Stolen Through Fraud in 2022, with Nearly 80 Per Cent of APP Fraud Cases Starting Online," May. 10, 2023. [Online]. Available: [https://www.ukfinance.org.uk/news-and-insight/press-release/over-ps12-billion-stolen-through-fraud-in-2022-nearly-80-cent-app]. [Accessed: Nov. 20, 2023].

[2] P. Harmston, I. Adjei, and K. Masterton-Smith, (2023). "Are you ready to combat authorised push payment fraud?," KPMG, Jun. 2023. [Online]. Available: https://kpmg.com/uk/en/blogs/home/posts/2023/06/app-fraud-publication.html. [Accessed: Nov. 20, 2023].

[3] Payment Systems Regulator, (2023). "PSR confirms new requirements for App Fraud Reimbursement," [Online]. Available: https://www.psr.org.uk/news-and-updates/latest-news/news/psr-confirms-new-requirements-for-app-fraud-reimbursement/. [Accessed: Nov. 20, 2023].

[4] A. Thennakoon, C. Bhagyani, S. Premadasa, S. Mihiranga, and N. Kuruwitaarachchi, (2019). "Real-time credit card fraud detection using machine learning," 2019 9th International Conference on Cloud Computing, Data Science & Engineering (Confluence), 2019. doi:10.1109/confluence.2019.8776942

[5] K. Searle and Sopra Steria, (2023), "Optimisation model for investigating fraudulent transactions in the banking sector." Edinburgh, Nov. 2023

[6] Panagiotopoulos, A. (2011), Optimising Time Series Forecasts Through Linear Programming.[PhD] University of Nottingham

[7] Zhou, C., Sun, C., Liu, Z., & Lau, F. (2015). A C-LSTM neural network for text classification. arXiv preprint arXiv:1511.08630.

[8] Benchaji, I., Douzi, S., El Ouahidi, B. et al., Enhanced credit card fraud detection based on attention mechanism and LSTM deep model. J Big Data 8, 151 (2021). https://doi.org/10.1186/s40537-021-00541-8

[9] J.-Q. Wang, P. Lu, H.-Y. Zhang, X.-H. Chen, (2014). "Method of multi-criteria group decision-making based on cloud aggregation operators with linguistic information". *Information Sciences*, vol. 274, Aug. 2014, pp. 177–91. ScienceDirect, https://doi.org/10.1016/j.ins.2014.02.130.

[10] J.-L., Marichal, (2009). Aggregation Functions for Decision Making. In *Decision-making Process* (eds D. Bouyssou, D. Dubois, M. Pirlot and H. Prade). https://doi.org/10.1002/9780470611876.ch17

[11] Y.-M. Wang, C. Parkan, Y. Luo, (2008), 'A Linear Programming Method for Generating the Most Favorable Weights from a Pairwise Comparison Matrix'. *Computers & Operations Research*, vol. 35, no. 12, Dec. 2008, pp. 3918–30. ScienceDirect, https://doi.org/10.1016/j.cor.2007.05.002.

A Appendix

A.1 Model 2 Stage 1

$$\min \quad \sum_i (1 - \text{combined}_i) \cdot s_i \cdot a_i + \sum_{d,b,i} \text{ext}_{d,b,i} \cdot E_i$$

$$\begin{aligned}
\text{s.t.} \quad \text{ext}_{d,b,i} &= \sum_{\substack{i \text{ s.t.} \\ \text{DD}_i = d, \\ \text{PP}_i = p}} \text{combined}_i \cdot \text{bt}_{i,b} && \forall d \in D, b \in B, p \in P && \text{(C1)} \\
\text{exp}_{d,b,i} &\leq \text{int}_{d,b,i} + \text{ext}_{d,b,i} && \forall d \in D, b \in B, is.t. p = PP_i && \text{(C2)} \\
\sum_p \text{int}_{d,b,p} &= \text{Z}_b && \forall d \in D, b \in B && \text{(C3)} \\
\text{combined}_i &\in [0,1] && \forall i \in I && \text{(C4)} \\
w^{tp}, w^{cp}, w^{start}, w^{mid}, w^{end}, w^a &\in [0,1] && \forall i \in I && \text{(C5)} \\
\boldsymbol{w^c} &\in (-\infty, 0.9] && \forall c \in C && \text{(C6)}
\end{aligned}$$

A.2 Model 2 - Example Results

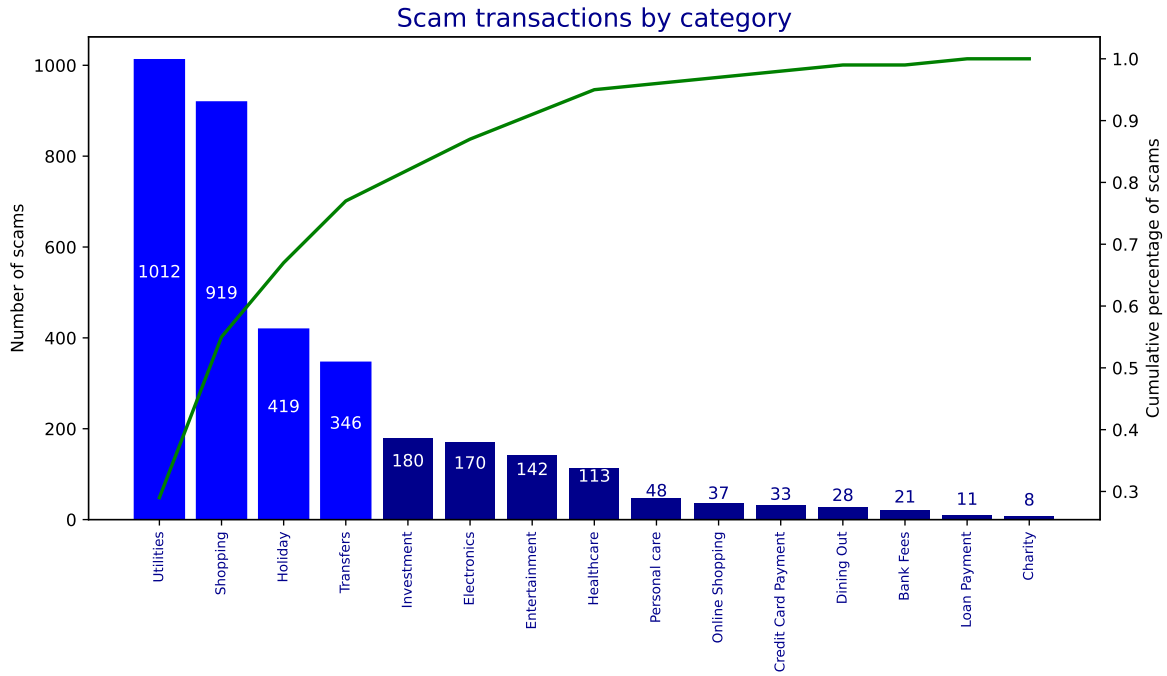
Table 16 shows the results for the weights found from the first iteration of stage 1 of model 2; these weights are computed based on 1st October 2023 to 31st July 2024, which are then applied to find the probability that each case on 1st August 2024 should be investigated.

Category weights ($\boldsymbol{w^c}$)	Bank Fees	-0.07233849
	Charity	-0.07233849
	Credit Card Payment	-0.07233849
	Dining Out	-0.07233849
	Electronics	-0.07233849
	Entertainment	0.9
	Groceries	-0.07233849
	Healthcare	-0.07233849
	Holiday	0.9
	Home Improvement	-0.07233849
	Housing	-0.07233849
	Investment	0.9
	Loan Payment	-0.07233849
	Online Shopping	-0.07233849
	Personal Care	0.9
	Shopping	0.9
	Streaming Services	-0.07233849
	Transfers	-0.07233849
	Transportation	-0.07233849
	Utilities	0.05832641
Other weights	Weight on amount (w^a)	0.0
	Weight on customer probability (w^{cp})	0.0
	Transaction probability (w^{tp})	0.0
	Weight on start of month (w^{start})	0.07233849056390607
	Weight on mid of month (w^{mid})	0.09999999999999998
	Weight on end of month (w^{end})	0.09999999999999998

Table 16: Weights from the first iteration of stage 1 of model 2.

A.3 Data Analysis: Scams by Category

Figure 5: Amount and proportion of fraudulent transactions in each of the 15 categories which contain frauds



A.4 Nomenclature Model 1

a_i	transaction value of the case i	
B	set of banks	$\{A, B, C, D, E\}$
C	set of categories (online shopping, utilities, housing, shopping, transfers transfers, transportation, dining out, groceries, electronics, streaming services, credit card payment, healthcare, bank fees, charity, home improvement, loan payment, holiday, investment, entertainment, personal care)	
cp_i	customer probability	$[0.2, 0.9]$
D	set of days of the month	
dp_i	description probability	$[0.1, 0.9]$
E_p	cost of hiring an external investigator for priority p	
I	set of all transactions	
m	constant to improve the fit	
q_i	adjusted probability of case i being a scam	
s_i	actual value, 1 if the case i is a scam, 0 otherwise	$\{0, 1\}$
$\hat{s}$	predicted value of whether transaction i is a scam, closer to 1 if it is more likely that it is fraudulent	
T_i	time it takes to investigate case i with priority p	
tp_i	transactional probability	$[0.1, 0.8]$
u_i	1 if the investigation of i will be done by one bank only	
$v_{i,bf}$	equals 1 if the case should be investigated and 0 otherwise	
$\boldsymbol{w}^b$	weight on whether there were 1 or 2 banks included in the transaction	
$\boldsymbol{w}^c$	weight on a given category k	
$\boldsymbol{w}^d$	weight on a given day of the month, d	
w^{dp}	weight on the description probability	
$\boldsymbol{w}^p$	weight on a given priority p of the case i	
w^{cp}	weight on the customer probability	
w^{tp}	weight on the transactional probability	
$\boldsymbol{x}_i^b$	value equal to 1 when 2 banks are included in the transaction	
$\boldsymbol{x}_i^c$	1 if category equals the category of case i	$\{0, 1\}$
$\boldsymbol{x}_i^d$	1 if day equals the day of the month when transaction i happened	$\{0, 1\}$
$\boldsymbol{x}_i^p$	1 if priority equals the priority of case i	$\{0, 1\}$
$y_{i,bf}$	1 if the case should hire an external investigator and 0 otherwise	
Z_b	size of investigation team at bank b	$(8, 12, 10, 10, 10)$

A.5 Nomenclature Model 2

a_i	transaction value of the case i	
a_f	transaction value of the case f, used in stage 2	
B	set of banks	$\{A, B, C, D, E\}$
$Btime_{i,b}$	time taken for bank b if it were to investigate case i given its priority	
$Btime_{f,b}$	1 if the bank b is involved in transaction f, used in stage 2	$\{0, 1\}$
C	set of categories (online shopping, utilities, housing, shopping, transfers transportation, dining out, groceries, electronics, streaming services, credit card payment, healthcare, bank fees, charity, home improvement, loan payment, holiday, investment, entertainment, personal care)	
$combined_i$	combined probability that a bank would be interested in investigating the case i, used to sample case in stage 2	
cp_i	customer probability of the user sending money in transaction i	$[0.2, 0.9]$
D	set of days of transactions	
DD_i	day of the case i	
E_i	cost of an external investigator for a transaction i with priority p	$(40, 60, 100, 150)$
E_f	cost of an external investigator for a transaction f with priority p, used in stage 2	$(40, 60, 100, 150)$
$exp_{d,b,i}$	expected time spent on a case i given its priority on day d by bank b	
$ext_{d,b,i}$	expected amount of time spent by external investigators on day d, at bank b, for case i	
$exttwo_f$	1 if the case f is investigated externally, 0 otherwise; used in stage 2	$\{0, 1\}$
F	set of candidate transactions in the stage 2	
I	set of all transactions	
$int_{d,b,p}$	expected amount of time spent by internal investigators on day d, at bank b, for cases of priority p	
$inttwo_{f,b}$	proportion of internal investigators contributed by bank b for case f, used in stage 2	$\{0, \frac{1}{2}, 1\}$
J	set of all transactions to sample from, stage 2	
$l_{i,c}$	equals 1 if case i has category c, 0 otherwise	$\{0, 1\}$
me_i	equals 1 if case i is completed after the 26th day of a month	$\{0, 1\}$
mm_i	equals 1 if case i is completed on the 5-26th day of a month	$\{0, 1\}$
ms_i	equals 1 if case i is completed up to the 4th day of a month	$\{0, 1\}$
P	set of priorities of transactions	$\{1, 2, 3, 4\}$
P	set of priorities of transactions	$\{1, 2, 3, 4\}$
s_i	1 if the case is a scam, 0 otherwise	$\in \{0, 1\}$
$sh_{i,b}$	1 when a case i is shared and bank b is involved, 0 otherwise	$\{0, 1\}$
$so_{i,b}$	1 if only one bank b could be involved in the review, 0 otherwise	$\{0, 1\}$
T_i	time spent for investigating a transaction i with priority p	$(0.25, 0.5, 1, 2)$
T_f	time to investigate case f given associated priority, used in stage 2	$(0.25, 0.5, 1, 2)$
tp_i	transactional probability of case i	$[0.1, 0.8]$
w^a	weight on the transaction amount	
$\boldsymbol{w^c}$	weight on a given category c	
w^{cp}	weight on the customer probability	
w^{start}	weight on transaction completed in the first 4 days of the month	
w^{mid}	weight on transaction completed between 5th - 26th of the month	
w^{end}	weight on transaction completed after the 26th of the month	
w^{tp}	weight on the transactional probability	
y_f	1 if the case f is investigated, 0 otherwise, used in stage 2	$\{0, 1\}$
Z_b	team size at the bank b	$(8, 12, 10, 10, 10)$